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$$f(x, y) = \begin{cases} \frac{x^3 - y^3}{x^2 + y^2} & \text{als } (x, y) \neq (0, 0) \\ 0 & \text{als } (x, y) = (0, 0) \end{cases}$$

Continuïteit:

Stel $x = r \cos \theta$ en $y = r \sin \theta$

$$\begin{aligned} \lim_{(x,y) \rightarrow (0,0)} f(x, y) &= \lim_{r \rightarrow 0} \frac{r^3 \cos^3 \theta - r^3 \sin^3 \theta}{r^2 \cos^2 \theta + r^2 \sin^2 \theta} \\ &= \lim_{r \rightarrow 0} \frac{r^3 (\cos^3 \theta - \sin^3 \theta)}{r^2} = \lim_{r \rightarrow 0} r (\cos^3 \theta - \sin^3 \theta) = 0 \end{aligned}$$

$\Rightarrow f$ is continu in $(0, 0)$

Partiële afgeleiden:

$$\begin{cases} \frac{\partial f}{\partial x}(0, 0) = \lim_{\lambda \rightarrow 0} \frac{f(\lambda, 0) - f(0, 0)}{\lambda} = \lim_{\lambda \rightarrow 0} \frac{\lambda^3}{\lambda^2} = \lim_{\lambda \rightarrow 0} \frac{\lambda}{\lambda} = 1 \\ \frac{\partial f}{\partial y}(0, 0) = \lim_{\lambda \rightarrow 0} \frac{f(0, \lambda) - f(0, 0)}{\lambda} = \lim_{\lambda \rightarrow 0} \frac{-\lambda^3}{\lambda^2} = \lim_{\lambda \rightarrow 0} \frac{-\lambda}{\lambda} = -1 \end{cases}$$

$\Rightarrow f$ is partieel afleidbaar in $(0, 0)$

Afleidbaarheid:

Stel $\bar{h} = (h_1, h_2)$

$$\begin{aligned} Df(\bar{0}, \bar{h}) &= \lim_{\lambda \rightarrow 0} \frac{f(\lambda h_1, \lambda h_2) - f(0, 0)}{\lambda} \\ &= \lim_{\lambda \rightarrow 0} \frac{\lambda^3 h_1^3 - \lambda^3 h_2^3}{\lambda^2 h_1^2 + \lambda^2 h_2^2} \\ &= \lim_{\lambda \rightarrow 0} \frac{\lambda^3 (h_1^3 - h_2^3)}{\lambda (\lambda^2 h_1^2 + \lambda^2 h_2^2)} \\ &= \lim_{\lambda \rightarrow 0} \frac{\lambda^3 (h_1^3 - h_2^3)}{\lambda^3 (h_1^2 + h_2^2)} = \frac{h_1^3 - h_2^3}{h_1^2 + h_2^2} \end{aligned}$$

$\Rightarrow f$ is afleidbaar in $(0, 0)$

Differentieerbaarheid:

$$\text{Voor differentieerbaarheid in } (0, 0) \text{ moet gelden: } Df(\bar{0}, \bar{h}) = \begin{pmatrix} \frac{\partial f}{\partial x}(0, 0) & \frac{\partial f}{\partial y}(0, 0) \end{pmatrix} \begin{pmatrix} h_1 \\ h_2 \end{pmatrix}$$

$$\begin{pmatrix} 1 & -1 \end{pmatrix} \begin{pmatrix} h_1 \\ h_2 \end{pmatrix} = h_1 - h_2$$

$$Df(\bar{0}, \bar{h}) = \frac{h_1^3 - h_2^3}{h_1^2 + h_2^2} \neq h_1 - h_2$$

$\Rightarrow f$ is niet differentieerbaar in $(0, 0)$

References